File Permission

Open permission.c /mnt/c/WINDOWS/system32/os/practical

```c
#include <stdio.h>
int main()
{
 FILE *fp;
 fp = fopen("sample.txt", "w");
 if (fp == NULL)
 {
 printf("Cannot open file\n");
 return 1;
 }
 fprintf(fp, "Hello, this is a test file.\n");
 fclose(fp);
 printf("File created successfully.\n");
 return 0;
}
```

1) Read and Write only to the owner

```
PS C:\WINDOWS\system32> wsl
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32 $ cd os
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os $ cd practical
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ./permission
File created successfully.
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ls -l sample.txt
-rwxrwxrwx 1 kaushik kaushik 28 Aug 22 02:54  sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ chmod 666 sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ls -l sample.txt
-rwxrwxrwx 1 kaushik kaushik 28 Aug 22 02:54  sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $
```

2) Read and Write to the owner and group

```
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ chmod 660 sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ls -l sample.txt
-rwxrwxrwx 1 kaushik kaushik 28 Aug 22 02:54  sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $
```

3) Only owner can read and write

```
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ chmod 600 sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ls -l sample.txt
-rwxrwxrwx 1 kaushik kaushik 28 Aug 22 02:54  sample.txt
```

4) Owner read/write, everyone else read-only

```
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ chmod 644 sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ls -l sample.txt
-rwxrwxrwx 1 kaushik kaushik 28 Aug 22 02:54  sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $
```

5) Owner can do everything, others can only read and execute

```
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ chmod 755 sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ls -l sample.txt
-rwxrwxrwx 1 kaushik kaushik 28 Aug 22 02:54  sample.txt
```

6) Owner and group can read/write

```
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ chmod 660 sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ls -l sample.txt
-rwxrwxrwx 1 kaushik kaushik 28 Aug 22 02:54  sample.txt
```

7) Only owner can execute

```
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ chmod 100 sample.txt
kaushik@LAPTOP-ASUSTUF3107 :/mnt/c/WINDOWS/system32/os/practical $ ls -l sample.txt
-r-xr-xr-x 1 kaushik kaushik 28 Aug 22 02:54  sample.txt
```

8) Owner: read + write; Group: nothing; Others: nothing → 600

9) Owner: read + write; Group: read; Others: read → 644

10) Owner: read + write + execute; Group: read + execute; Others: read + execute → 755

11) Owner: read + write; Group: read + write; Others: nothing → 660

12) Everyone: read only → 444